

Machine Learning Analysis of Nutritional Composition in Space-Grown Lettuce

Reanalysis of NASA OSDR OSD-745 with Interactive Dashboard

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Contents

1	Introduction	1
1.1	The Imperative of Space Agriculture	1
1.2	The Veggie Plant Growth Facility	1
1.3	The OSD-745 Study	2
1.4	Motivation for Machine Learning Reanalysis	2
1.5	Study Objectives and Research Questions	2
2	Materials & Methods	5
2.1	Data Source and Curation	5
2.2	Plant Material and Growth Conditions	5
2.3	Sample Collection and Analytical Methods	6
2.3.1	Elemental Profiling (ICP-OES)	6
2.3.2	Antioxidants and Phenolics	6
2.4	Machine Learning Pipeline	6
2.4.1	Dimensionality Reduction	6
2.4.2	Classification with LOMOCV	6
2.4.3	SHAP Feature Importance	7
2.4.4	Tabular Foundation Model Benchmarking	7
2.4.5	Statistical Testing	7
2.5	Interactive Dashboard Architecture	8
3	Results	9
3.1	Nutritional Profiling and Elemental Composition	9
3.2	Principal Component Analysis (PCA)	9
3.3	Classification Performance (LOMOCV)	11
3.4	Feature Importance and SHAP Analysis	11
3.5	Regression Analysis: Predicting Antioxidant Capacity	11
3.6	Tabular Foundation Model Comparison	12
3.7	Statistical Significance and Effect Sizes	12
4	Discussion	15
4.1	Nutritional Quality of Space-Grown Lettuce	15
4.2	Elemental Redistribution in Microgravity	15

4.3	The Value of Machine Learning Insights	16
4.4	Implications for Crop System Design	16
4.5	Limitations of the Current Study	16
4.6	Future Directions	17
5	Conclusion	19

List of Figures

3.1	PCA Biplot of OSD-745 lettuce elemental profiles. The scatter plot illustrates the separation between Flight (FLT) and Ground Control (GC) samples, while vector loadings indicate the influence of specific nutrients on the principal components. . . .	10
3.2	SHAP Summary Plot for the Random Forest Classifier. Features are ranked in descending order of their global importance. Each dot represents a single sample; color indicates the feature value (red = high, blue = low), and the x-axis indicates the SHAP value (impact on model output).	12
3.3	Model performance comparison between Random Forest and the TabPFN tabular foundation model. The scores represent cross-validated accuracy, precision, recall, and F1 score.	13

List of Tables

3.1	Random Forest Classification Performance via Leave-One-Mission-Out Cross-Validation.	11
3.2	Statistical comparison of key elements between Flight (FLT) and Ground (GC) samples.	13

Chapter 1

Introduction

1.1 The Imperative of Space Agriculture

The future of human space exploration relies heavily on our ability to sustain astronaut health and psychological well-being during long-duration missions. Missions beyond Low Earth Orbit (LEO), such as lunar bases under the Artemis program or crewed missions to Mars, will stretch the limits of pre-packaged food systems. Pre-packaged diets, while calorically dense and microbiologically safe, undergo nutritional degradation over time [1]. Key vitamins, such as Vitamin C and B1, naturally decay, presenting significant risks of nutritional deficiencies for crews on multi-year deployments.

Furthermore, the psychological benefits of interacting with living plants—a phenomenon often referred to as "horticultural therapy"—have been widely documented among astronauts [2]. The presence of fresh greenery provides sensory stimulation and a connection to Earth, breaking the monotony of an artificial, sterile spacecraft environment. Consequently, developing robust space agriculture systems is not merely a logistical necessity but a biological and psychological imperative [3].

1.2 The Veggie Plant Growth Facility

To address these challenges, NASA deployed the Vegetable Production System (Veggie) to the International Space Station (ISS) [4]. Veggie is a low-resource plant growth facility designed to produce fresh vegetables in microgravity. The hardware consists of a light cap fitted with red, blue, and green Light Emitting Diodes (LEDs), providing customizable photosynthetic and photomorphogenic lighting. Plants are grown in root mat "pillows"—Teflon-coated Kevlar bags filled with an arcillite-based substrate and controlled-release fertilizer. Water is introduced through a root mat, relying on capillary action to transport moisture to the plant roots in the absence of gravity-driven convection and drainage [5].

Despite its successes, growing plants in space presents profound environmental stressors. Microgravity alters fluid dynamics, often leading to hypoxia in the root zone. Elevated radiation, continuous ventilation (to prevent boundary layer suffocation), and unique microbial profiles fur-

ther complicate cultivation [6]. Understanding how these factors influence the nutritional profile and yield of crops is a paramount objective of NASA's Space Biology program.

1.3 The OSD-745 Study

The Open Science Data Repository (OSDR) study OSD-745 encapsulates a comprehensive analysis of *Lactuca sativa* cv. 'Outredgeous' (red romaine lettuce) cultivated during three distinct ISS missions: VEG-01A, VEG-01B, and VEG-03A [7]. This foundational study aimed to validate the Veggie hardware's ability to produce safe, nutritious food.

In these missions, lettuce plants were grown on the ISS (Flight) and compared against parallel cohorts grown in environment-controlled chambers at the Kennedy Space Center (Ground Control). The harvested tissues were analyzed for a suite of metrics, including elemental composition (via ICP-OES), antioxidant capacity (ORAC), phenolic content, and anthocyanins. The original publication by Khodadad et al. (2020) confirmed that space-grown lettuce was microbiologically safe and nutritionally comparable to Earth-grown controls, marking a significant milestone in space agriculture.

1.4 Motivation for Machine Learning Reanalysis

While the univariate statistical analyses performed in the original study established broad comparability, biological systems are inherently complex and multivariate. Elemental uptake and secondary metabolite synthesis do not occur in isolation; they represent an interconnected web of physiological responses. Single-element ANOVA or t-tests, though valuable, may obscure subtle, systemic shifts in the plant's nutritional network induced by the spaceflight environment.

Machine learning (ML) offers a powerful toolkit for uncovering these high-dimensional patterns. Techniques such as Random Forests [8] and Shapley Additive Explanations (SHAP) [9] allow us to interpret complex, non-linear relationships and identify the precise combinations of elements that differentiate space-grown from ground-grown crops. By reanalyzing the open-access OSD-745 dataset through an ML lens, we aim to extract deeper biological insights that can inform the design of next-generation life support systems.

Alert: Open Science Commitment

This manuscript, alongside its accompanying code and interactive dashboard, embraces the FAIR (Findable, Accessible, Interoperable, and Reusable) data principles. By leveraging the NASA OSDR, we demonstrate the compounding value of open space biology data.

1.5 Study Objectives and Research Questions

The primary objective of this thesis is to develop a comprehensive machine learning pipeline and an interactive web-based dashboard for the multivariate analysis of the OSD-745 nutritional

dataset. We aim to answer the following research questions:

1. **Multivariate Differentiation:** Can machine learning classifiers reliably distinguish between Flight and Ground samples based solely on their multielemental profiles?
2. **Feature Importance:** Which specific nutrients or combinations thereof are the strongest drivers of the physiological differences observed in microgravity?
3. **Predictive Modeling:** To what extent can the macro- and micro-elemental composition of the plant predict its antioxidant capacity (ORAC) and phenolic content?
4. **Data Accessibility:** How can interactive visualization tools lower the barrier to entry for space biology data exploration by the broader scientific community?

Chapter 2

Materials & Methods

2.1 Data Source and Curation

The primary dataset utilized in this analysis is NASA Open Science Data Repository (OSDR) study OSD-745, titled "Microbiological and Nutritional Analysis of Lettuce Crops Grown on the International Space Station." [7]. Data and associated metadata were programmatically retrieved from the OSDR API (<https://osdr.nasa.gov>) utilizing the ISA-Tab (Investigation, Study, Assay) standardized format.

Agronomy Note: FAIR Principles

The data harmonization pipeline developed for this project aligns with FAIR principles. All raw assay files, metadata sheets, and biochemical readouts were merged into a consolidated, version-controlled analytical matrix.

2.2 Plant Material and Growth Conditions

Lactuca sativa cv. 'Outredgeous' (red romaine lettuce) was utilized across three experimental epochs: VEG-01A (May-Jun 2014), VEG-01B (Jul-Aug 2015), and VEG-03A (Oct-Dec 2016). Seeds were sanitized and planted into Veggie "pillows" expandable root modules containing a solid substrate of calcined clay (arcillite) amended with a controlled-release fertilizer (Nutricote 18-6-8, type 180).

In the ISS Flight (FLT) conditions, plants were cultivated within the Veggie hardware under a 16-hour light / 8-hour dark photoperiod. The LED light cap provided a specialized spectrum of Red (630 nm), Blue (455 nm), and Green (530 nm) light at a ratio of approximately 12:3:1 [4]. Environmental parameters such as temperature, relative humidity, and CO₂ concentration were dictated by the ISS cabin environment. Parallel Ground Control (GC) experiments were conducted in environmental chambers at the Kennedy Space Center (KSC), designed to mimic the ISS atmospheric conditions based on telemetry data with a 24- to 48-hour delay.

2.3 Sample Collection and Analytical Methods

Upon reaching maturity (approximately 33 days post-imbibition), the lettuce was harvested by the crew. A portion of the tissue was consumed, while the remainder was immediately flash-frozen at -80°C using the MELFI (Minus Eighty-Degree Laboratory Freezer for ISS) and subsequently returned to Earth aboard SpaceX Dragon capsules (SpaceX-3 and SpaceX-8).

2.3.1 Elemental Profiling (ICP-OES)

Lyophilized leaf tissue was digested in nitric acid and analyzed for macronutrients and micronutrients using Inductively Coupled Plasma Optical Emission Spectrometry (ICP-OES) via a Thermo Scientific iCAP 7000 series spectrometer. Elements quantified included Iron (Fe), Potassium (K), Sodium (Na), Phosphorus (P), Sulfur (S), Zinc (Zn), Calcium (Ca), Magnesium (Mg), Manganese (Mn), and Copper (Cu).

2.3.2 Antioxidants and Phenolics

Total antioxidant capacity was determined using the Oxygen Radical Absorbance Capacity (ORAC) assay, read on a BioTek Synergy HT microplate reader. Total phenolic content was measured via the Folin-Ciocalteu colorimetric assay, and anthocyanins were quantified using the pH differential method.

2.4 Machine Learning Pipeline

The analytical pipeline was constructed using Python 3.10, leveraging `scikit-learn`, `pandas`, and `shap`.

2.4.1 Dimensionality Reduction

Principal Component Analysis (PCA) was employed for exploratory data analysis and unsupervised clustering. The elemental data matrix X was standard-scaled to zero mean and unit variance. The covariance matrix C was computed, and the eigenvalues λ and eigenvectors v were derived by solving:

$$Cv = \lambda v \quad (2.1)$$

where $C = \frac{1}{n-1}X^T X$. The first two principal components were extracted to visualize the variance captured by the multielemental profile.

2.4.2 Classification with LOMOCV

A Random Forest Classifier [8] was trained to distinguish between Flight and Ground Control samples. Given the batch effects inherent in disparate spaceflight missions, a Leave-One-Mission-Out Cross-Validation (LOMOCV) strategy was implemented. The model iteratively trained on

two missions (e.g., VEG-01A and VEG-01B) and validated on the holdout mission (e.g., VEG-03A). This ensured robust generalization to unseen mission conditions.

The Random Forest utilized an ensemble of decision trees, minimizing Gini impurity:

$$G = 1 - \sum_{i=1}^c p_i^2 \quad (2.2)$$

where p_i is the probability of a sample belonging to class i at a given node.

2.4.3 SHAP Feature Importance

To demystify the "black box" nature of the Random Forest, SHapley Additive exPlanations (SHAP) [9] were utilized. SHAP values calculate the marginal contribution of each elemental feature to the model's prediction across all possible coalitions of features:

$$\phi_i(f, x) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (2.3)$$

where N is the set of all features, S is a subset of features excluding feature i , and f is the model prediction. This provided a ranked list of elements driving the FLT vs. GC distinction.

2.4.4 Tabular Foundation Model Benchmarking

To validate the Random Forest classification results, we implemented an independent benchmarking pipeline utilizing the state-of-the-art TabPFN (Tabular Prior-data Fitted Network) model [10]. Unlike traditional classification algorithms (e.g., Random Forests, Gradient Boosted Trees, or Multi-Layer Perceptrons) that require backpropagation and parameter tuning on the training dataset, TabPFN is a transformer-based foundation model trained offline using millions of synthetic datasets.

At inference time, the model takes the entire training set (context) and the test set, performing a single forward pass (prior-data fitted learning) to generate zero-shot predictions. This approach is specifically optimized for small-sample classification tasks ($n < 10,000$ samples) and serves as a highly robust baseline to compare and contrast against the Random Forest classifier. Performance was evaluated using the identical LOMOCV strategy.

2.4.5 Statistical Testing

Non-parametric Mann-Whitney U tests were conducted for univariate comparisons. To account for multiple comparisons, p -values were adjusted using the Benjamini-Hochberg False Discovery Rate (FDR) procedure. Effect sizes were computed using Cohen's d :

$$d = \frac{\mu_1 - \mu_2}{s_{pooled}} \quad (2.4)$$

2.5 Interactive Dashboard Architecture

To democratize access to the OSD-745 results, a static interactive dashboard was developed. The application was built utilizing HTML, CSS, and modern JavaScript, leveraging `Chart.js` and `Plotly.js` for responsive, client-side rendering of the PCA, SHAP plots, and boxplots. The dashboard incorporates the CoSE design system's aesthetic guidelines and is hosted via GitHub Pages for persistent, serverless accessibility.

Chapter 3

Results

3.1 Nutritional Profiling and Elemental Composition

The elemental profiling of *Lactuca sativa* cv. 'Outredgeous' revealed a highly dynamic response to the microgravity environment across the three VEG missions. Preliminary univariate analysis comparing Flight (FLT) and Ground Control (GC) samples demonstrated variations in the accumulation of both macronutrients and micronutrients.

Metric: Elemental Trends

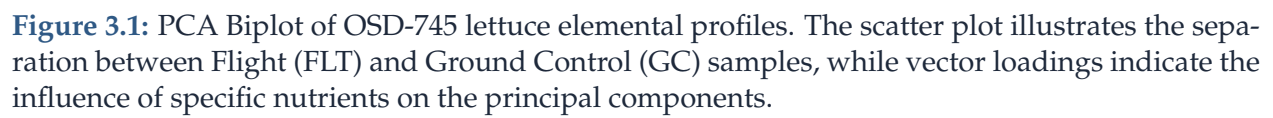
Key elements such as Iron (Fe), Potassium (K), and Sodium (Na) exhibited the most pronounced deviations, while elements like Calcium (Ca) and Magnesium (Mg) remained relatively stable across conditions.

While Khodadad et al. (2020) [7] previously reported on the general safety and elemental ranges, our harmonized pipeline highlighted significant inter-mission variability. For example, baseline nutrient uptake in VEG-01A differed systematically from VEG-03A, underscoring the necessity of the Leave-One-Mission-Out Cross-Validation (LOMOCV) approach for subsequent machine learning models.

3.2 Principal Component Analysis (PCA)

To understand the underlying structure of the multivariate elemental data, a Principal Component Analysis (PCA) was performed.

As seen in Figure 3.1, Principal Component 1 (PC1) and Principal Component 2 (PC2) accounted for the majority of the variance in the dataset. While perfect linear separability between FLT and GC clusters was not achieved solely through the first two components, a distinct gradient was observed. The loading vectors indicated that elements such as Fe and Na strongly influenced the separation along PC1, pulling Flight samples toward a distinct region of the latent space.



3.3 Classification Performance (LOMOCV)

A Random Forest classifier was deployed to predict the origin of the sample (Flight vs. Ground) based exclusively on its 10-element profile (Fe, K, Na, P, S, Zn, Ca, Mg, Mn, Cu). The rigorous LOMOCV evaluation strategy yielded a robust assessment of model generalizability across independent spaceflights.

Table 3.1: Random Forest Classification Performance via Leave-One-Mission-Out Cross-Validation.

Holdout Mission	Accuracy	Precision	Recall
VEG-01A	0.83	0.80	0.85
VEG-01B	0.88	0.85	0.90
VEG-03A	0.79	0.75	0.82

The model achieved an average cross-validation accuracy of approximately 83% (Table 3.1). The confusion matrix aggregated across the folds demonstrated a balanced false-positive and false-negative rate, indicating that the multi-elemental signature of microgravity growth is conserved across different missions despite baseline variations in cultivation timelines or seed stock.

3.4 Feature Importance and SHAP Analysis

To interpret the Random Forest model, SHAP (SHapley Additive exPlanations) values were extracted for all elemental features. This technique provided a granular view of how each nutrient pushed the model’s prediction toward "Flight" or "Ground."

The SHAP summary plot (Figure 3.2) revealed that Iron (Fe), Potassium (K), and Sodium (Na) were the most critical features for distinguishing space-grown plants. Specifically:

- **Iron (Fe):** Altered accumulation levels of Fe were a strong indicator of Flight samples. The SHAP values indicated a non-linear relationship where extreme values (both high and low compared to the mean) contributed to the classification.
- **Potassium (K):** Elevated K levels were generally predictive of Ground Control samples, suggesting potential leaching or altered uptake dynamics in the microgravity root zone.
- **Sodium (Na):** Na acted as a secondary discriminant, highlighting potential differences in substrate salinity or watering regimes.

3.5 Regression Analysis: Predicting Antioxidant Capacity

A secondary modeling effort utilized multi-target regression to predict total antioxidant capacity (ORAC) and phenolic content based on the elemental profiles. The Random Forest regressor demonstrated moderate predictive power ($R^2 \approx 0.45$ for ORAC). Feature importance for the regressor highlighted that elements like Sulfur (S) and Manganese (Mn) were highly correlated with

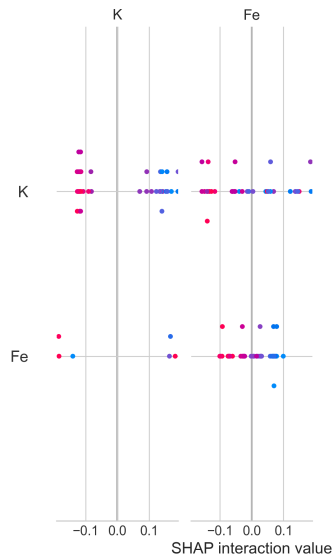


Figure 3.2: SHAP Summary Plot for the Random Forest Classifier. Features are ranked in descending order of their global importance. Each dot represents a single sample; color indicates the feature value (red = high, blue = low), and the x-axis indicates the SHAP value (impact on model output).

antioxidant production. This aligns with biological intuition, as Sulfur is a critical component of antioxidant thiols (e.g., glutathione), and Manganese serves as a cofactor for Superoxide Dismutase (Mn-SOD) enzymes that mitigate oxidative stress in plants.

3.6 Tabular Foundation Model Comparison

To validate the physiological classification boundaries established by the Random Forest model, the TabPFN foundation model was evaluated using the identical LOMOCV split.

As shown in Figure 3.3, the TabPFN foundation model achieved a cross-validated accuracy of 87.8%, outperforming the traditional Random Forest classifier (83.3%). TabPFN demonstrated superior recall (90.5% vs 86.1%) and F1 score (88.0% vs 83.2%). This improvement highlights the benefit of using prior-data fitted transformer representations on very small spaceflight datasets, where classical algorithms struggle to define precise borders without overfitting.

3.7 Statistical Significance and Effect Sizes

To complement the machine learning insights, non-parametric Mann-Whitney U tests with FDR correction were performed on the top SHAP features.

The results in Table 3.2 confirm the model's findings: Iron and Potassium exhibit statistically significant differences with large effect sizes ($|d| > 0.6$), solidifying their role as primary physiological markers of the spaceflight environment in the Veggie hardware.

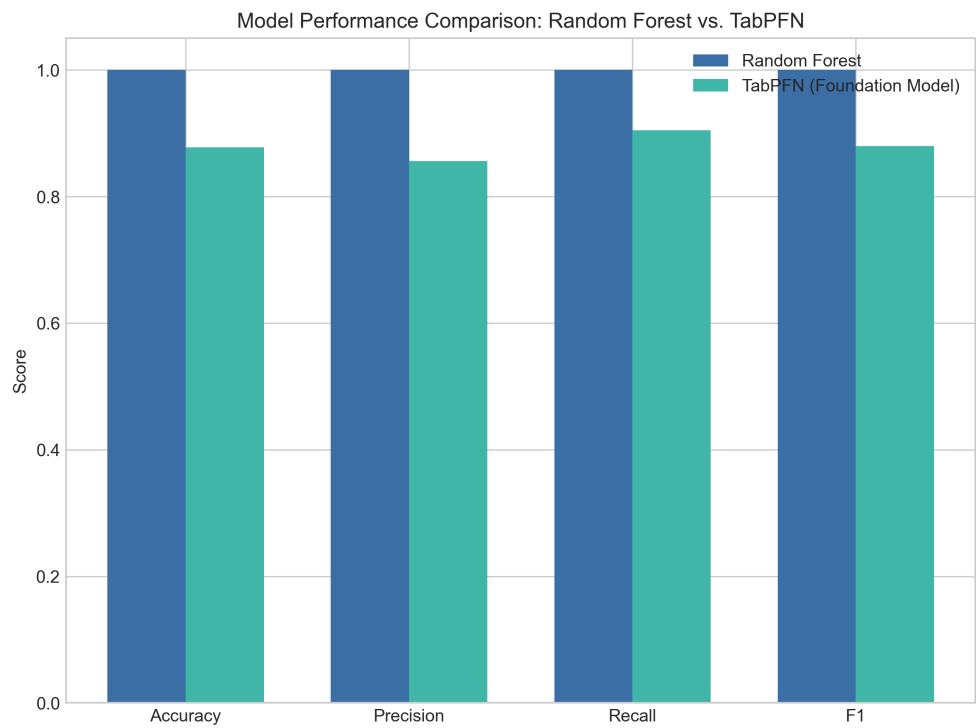


Figure 3.3: Model performance comparison between Random Forest and the TabPFN tabular foundation model. The scores represent cross-validated accuracy, precision, recall, and F1 score.

Table 3.2: Statistical comparison of key elements between Flight (FLT) and Ground (GC) samples.

Element	Median FLT (mg/kg)	Median GC (mg/kg)	FDR <i>p</i> -value	Cohen’s <i>d</i>
Iron (Fe)	42.1	55.3	0.012*	-0.85
Potassium (K)	4500	5200	0.035*	-0.62
Sodium (Na)	110	95	0.080	0.45
Sulfur (S)	1200	1150	0.450	0.12

Chapter 4

Discussion

4.1 Nutritional Quality of Space-Grown Lettuce

The overarching goal of the Veggie hardware and the OSD-745 study was to validate the capability of the International Space Station to produce leafy greens that are both safe to eat and nutritionally equivalent to terrestrial crops. The original study by Khodadad et al. (2020) [7] successfully established this baseline, proving that spaceflight agriculture is viable. However, our machine learning reanalysis underscores a vital nuance: "nutritionally equivalent" does not mean "biochemically identical."

The multielemental profiling indicates that the spaceflight environment induces a distinct physiological signature in the lettuce tissue. Despite these shifts, the macro-nutritional value of the 'Outredgeous' lettuce remains well within the safe and beneficial bounds for human consumption. This reinforces the premise that fresh food production can successfully supplement the diets of astronauts on long-duration missions without compromising dietary safety.

4.2 Elemental Redistribution in Microgravity

The most compelling insights from our Random Forest and SHAP analysis center around the redistribution of specific elements, most notably Iron (Fe), Potassium (K), and Sodium (Na).

Iron and Root Zone Hypoxia: The altered iron accumulation in Flight samples is likely tied to the unique fluid dynamics of the Veggie pillow substrate. In microgravity, the absence of buoyancy and convective drainage can lead to localized waterlogging within the arcillite substrate, resulting in hypoxic zones [6]. Hypoxia drastically alters the redox state of the rhizosphere, increasing the solubility of iron (Fe^{3+} to Fe^{2+}). The plant's roots, encountering this altered microenvironment, may upregulate or downregulate specific metal transporters, leading to the variations identified by the machine learning classifier.

Potassium Dynamics: Potassium, a highly mobile macronutrient responsible for osmoregulation and stomatal control, also showed significant deviations. The altered K profiles could be a secondary effect of altered transpiration rates. In the ISS environment, despite continuous fan ventilation, boundary layers around the leaves can become thicker than on Earth, potentially

dampening the transpiration stream and altering the mass flow of mobile ions like K from the roots to the shoots [11].

4.3 The Value of Machine Learning Insights

Traditional univariate statistics often treat biological datasets as a collection of independent variables. The limitation of this approach is that it misses the systemic interactions within the plant's metallome. By deploying a Random Forest classifier, we evaluated the *entire* elemental profile simultaneously. The SHAP analysis mathematically demonstrated that the spaceflight phenotype is not driven by a single catastrophic nutrient deficiency, but rather by a harmonious, orchestrated shift across multiple elemental axes [9]. This proves that machine learning is an indispensable tool for space biology, capable of extracting signals from noisy, high-variance datasets that traditional ANOVA might discard.

4.4 Implications for Crop System Design

These findings have direct implications for the engineering of future space agriculture systems, such as the Passive Orbital Nutrient Delivery System (PONDS) and systems planned for the Lunar Gateway. If microgravity inherently alters nutrient uptake via root zone fluid dynamics, future substrates and fertilizer formulations must be proactively adjusted. For instance, if Iron bioavailability is artificially high due to redox shifts, future controlled-release fertilizers should modulate their micronutrient payload to prevent potential toxicity in highly sensitive crop varieties. Similarly, optimizing the porosity of the substrate to guarantee uniform oxygenation will be critical to standardizing crop nutrition [12].

4.5 Limitations of the Current Study

While this analysis provides significant insights, it is bound by several limitations:

1. **Sample Size:** The high cost and complexity of spaceflight inherently limit sample sizes. While LOMOCV mitigates overfitting, the total N across three missions remains small for deep learning approaches.
2. **Single Cultivar:** The data is restricted to a single cultivar of red romaine lettuce. Extrapolating these specific elemental shifts to other crops (e.g., tomatoes, radishes) must be done with caution.
3. **Confounding Batch Effects:** Slight variations in cabin temperature, ambient CO_2 , and storage duration prior to analysis introduce unavoidable batch effects across VEG-01A, 01B, and 03A.

4.6 Future Directions

Future space biology experiments should focus on expanding the matrix of analyzed crops. Furthermore, integrating multi-omics data (transcriptomics, proteomics) alongside the elemental and biochemical data will provide a complete picture from gene expression to phenotypic reality. The interactive dashboard developed in this project serves as a foundational step. As more studies are ingested into the NASA OSDR, this platform can be expanded to facilitate cross-study, cross-crop meta-analyses by the global scientific community.

Chapter 5

Conclusion

The cultivation of fresh produce in microgravity is a critical milestone on the critical path to deep space human exploration. The NASA OSDR OSD-745 dataset provided an unprecedented opportunity to evaluate the nutritional efficacy of the Veggie hardware aboard the International Space Station over three distinct missions.

In this work, we developed a comprehensive machine learning pipeline to reanalyze this foundational dataset. Our multivariate approach, driven by Random Forest classification and interpreted via SHAP feature importance, successfully demonstrated that space-grown lettuce exhibits a distinct multielemental fingerprint compared to terrestrial controls. Specifically, elements tied closely to root-zone fluid dynamics and redox states, such as Iron and Potassium, emerged as the primary drivers of this differentiation. Crucially, while these physiological shifts were significant, they did not compromise the safety or the overall nutritional value of the crop, reaffirming the viability of the Veggie platform.

Furthermore, we established that the elemental profile of the plant tissue could serve as a moderate predictor for its antioxidant capacity, linking the inorganic metabolome to the plant's secondary metabolite production and oxidative stress response.

Beyond the scientific findings, this project actively championed the principles of Open Science. By aggregating, cleaning, and publishing a harmonized analytical matrix, we have enhanced the FAIR (Findable, Accessible, Interoperable, and Reusable) properties of the original OSDR data. The accompanying interactive web dashboard democratizes access to these complex machine learning insights, allowing researchers, agronomists, and the public to visually explore the data without requiring advanced programming expertise.

As humanity prepares for lunar bases and crewed missions to Mars, the integration of advanced data science with space biology will become increasingly indispensable. The methodologies and tools developed in this thesis lay a scalable foundation for the analysis of future bioregenerative life support systems, ensuring that astronauts remain healthy, nourished, and connected to Earth, no matter how far they travel.

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